

Computational Thinking and Python Basics

Learn how a computer follows explicit instructions and write a first Python program.



WHY THIS MATTERS

Start from what you already know

No prior coding knowledge

- 01 Explain algorithms in plain language.
- 02 Use variables in a small worked example.
- 03 Complete the guided lab and verify syntax errors.

CORE CONCEPT

The vocabulary of this topic

Learn how a computer follows explicit instructions and write a first Python program.

algorithms: An algorithm is a finite, ordered set of unambiguous steps that transforms an input into an output.

variables: A variable is a name bound to a value so a program can reuse or replace that value.

expressions: An expression combines values, variables, and operators to calculate a new value.

syntax errors: A syntax error means Python cannot parse the written instructions according to its grammar.

FOUR CONNECTED IDEAS

How the concept works

01

algorithms

02

variables

03

expressions

04

syntax errors

An input is stored in variables, transformed by expressions, and displayed as output.

CORE CONCEPT

Read every part of the mechanism

An input is stored in variables, transformed by expressions, and displayed as output.

variable: A named reference to a value.

expression: Values and operators that produce a result.

WORKED EXAMPLE

Worked example by hand

Store a course price of 8000 and a 10% discount; calculate and print 7200.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
price = 8000
discount_rate = 0.10
final_price = price - price * discount_rate
print(final_price)
```

OUTPUT

7200.0

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model



An algorithm must terminate and produce the same result for the same stated inputs.



``=`` assigns a value; ``==`` asks whether two values are equal.



Python evaluates ``*`` before ``+``; parentheses make intended order explicit.



Read an error from its final line, then inspect the reported file and line number.

FOUR CONNECTED IDEAS

Notebook readiness map

01

02

03

04

algorithms**variables****expressions****syntax errors**

Use the concepts from this lesson to read and complete
<https://github.com/curiously/Al-Bootcamp/blob/master/01.python-essentials-for-ai.ipynb>.

RISK REVIEW

Common beginner mistakes

01 Reading `=` as mathematical equality instead of assignment.

02 Treating an error message as a failure rather than diagnostic evidence.

GUIDED LAB

Write a program that converts study minutes into hours and remaining minutes.

01

02

03

04

Predict

Run

Inspect

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

- 01 Explain algorithms in plain language.
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NEXT EVIDENCE

A runnable Python file with named inputs, a calculation, output, and comments explaining each line.